# 28 Supersymmetric Versions of the Standard Model

## 28.7 Baryon and Lepton Non-Conservation

The extra particles in supersymmetric models provide several new mechanisms for baryon and lepton non-conservation. We saw in Section 28.1 that there are various baryon- and lepton-number non-conserving supersymmetric operators (28.1.2) and (28.1.3) of dimensionality four that can be included in a renormalizable $SU(3) \cdot SU(2) \cdot U(1)$-invariant theory and that would lead to processes like proton decay at a catastrophic rate. These terms can be excluded from the Lagrangian by imposing $R$ parity conservation (or, equivalently, invariance under a change of sign of all quark and lepton chiral superfields) but this does not exclude various $SU(3) \cdot SU(2) \cdot U(1)$-invariant but baryon- and lepton-number non-conserving operators of dimensionality $d > 4$. As discussed in Section 21.3, if there is an underlying mechanism for baryon and lepton non-conservation characterized by some high mass scale $M$, then these operators will appear in the effective Lagrangian of the standard model with coefficients proportional to $M^{4-d}$. With only the fields of the non-supersymmetric standard model, operators that can violate baryon conservation have a minimum dimensionality six [36], and thus give baryon non-conserving amplitudes proportional to $M^{-2}$. The new fields required by supersymmetry lead to two important changes in estimates of baryon non-conserving processes like proton and bound neutron decay. As we saw in Section 28.2, the change in the renormalization group equations gives larger estimates of $M$, which decreases the effect of the dimension six operators. At the same time, these new fields allow the construction of new operators of dimensionality five, which give baryon non-conserving amplitudes proportional to $M^{-1}$ and are therefore likely to make the dominant contribution to proton and bound neutron decay [37].

The supersymmetric operators of dimensionality five that can be formed out of chiral superfields (generically called $\Phi$) are of the forms

$$(\Phi^* \Phi\Phi)_D \quad \text{and} \quad (\Phi\Phi\Phi\Phi)_{\mathcal{F}} \, ,$$

and their complex conjugates. (We do not consider operators that include derivatives or gauge fields, because they do not offer any additional possibility of baryon or lepton non-conservation.)

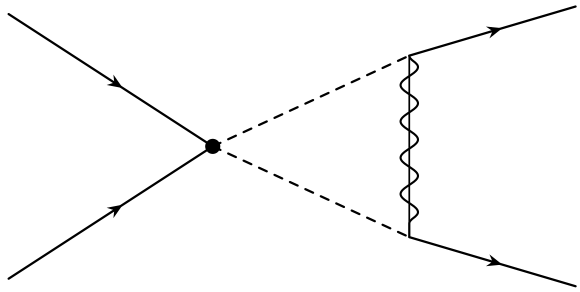

**Figure 28.7**. A diagram that can produce a four-fermion interaction among quarks and/or leptons that violates baryon and lepton number conservation. Here solid lines are quarks and/or leptons; dashed lines are squarks and/or sleptons; the combined solid and wavy line is a gaugino; and the dot is a vertex arising directly from the $\mathcal{F}$-term interactions (28.7.3).

In the notation of Section 28.1, the $SU(3)\cdot SU(2)\cdot U(1)$-invariant operators of dimensionality five that also conserve $R$ parity are

$$(LLH_2H_2)_{\mathcal{F}}\,, \tag{28.7.1}$$

$$(L\bar{E}H_2^*)_D\,, \qquad (Q\bar{D}H_2^*)_D\,, \qquad (Q\bar{U}H_1^*)_D\,, \qquad (QQ\bar{U}\bar{D})_{\mathcal{F}}\,, \qquad (Q\bar{U}L\bar{E})_{\mathcal{F}}\,, \tag{28.7.2}$$

and

$$(QQQL)_{\mathcal{F}}\,, \qquad (\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}\,, \tag{28.7.3}$$

with obvious contraction of indices as dictated by $SU(3)$ and $SU(2)$ conservation. The interaction (28.7.1) is the supersymmetric version of the dimensionality five operator which would provide small neutrino masses in some theories [38]. The interactions (28.7.2) only provide small corrections to processes that already occur in the renormalizable terms of the supersymmetric standard model. It is the interactions (28.7.3) that provide new mechanisms for baryon as well as lepton non-conservation.

According to Eq. (26.4.4), the quarks and leptons enter in the interactions (28.7.3) through terms involving a pair of quark and/or lepton fields and a pair of squark and/or slepton fields. In order to generate reactions among quarks and leptons alone, it is necessary for the pair of squarks and/or sleptons to be converted into a pair of quarks and/or leptons by exchanging a gaugino in the one-loop diagram shown in Figure 28.7. This will produce effective four-fermion $qqq\ell$ interactions with $d=6$ among three quarks and a lepton. The coupling $g_6$ of these interactions will be proportional to the square of the gauge coupling $g$ or $g'$ or $g_s$ of the gaugino, to the supersymmetry-breaking mass of the gaugino, to the inverse square of the larger of the gaugino and squark or slepton masses (which is needed to give a coupling of the right dimensionality), and to a factor of order $1/8\pi^2$ from the loop integration.

It might be thought that the greater strength of the gluino coupling would make gluino exchange the dominant contribution to $g_6$. (Indeed, in gauge-mediated supersymmetry-breaking theories, for moderate values of the trace (28.6.3) and with $g\approx g'$, Eqs. (28.6.13)–(28.6.14) give

$$m_{\rm gluino}\approx m_{\rm squark}\approx \frac{g_s^2}{16\pi^2}M_*\,, \qquad m_{\rm wino}\approx m_{\rm slepton}\approx m_{\rm bino}\approx \frac{g^2}{16\pi^2}M_*\,,$$

where $M_*$ is a mass characterizing the messenger sector. Therefore in such theories individual gluino exchange diagrams give contributions proportional to $g_s^2/m_{\rm gluino}$, while wino or bino exchange (or, more precisely, chargino or neutralino exchange) makes a contribution proportional to $g^2 m_{\rm wino}/m^2_{\rm squark}$, which is smaller by a factor roughly equal to $m_{\rm wino}g_s^2/m_{\rm gluino}g^2\approx g^4/g_s^4$.) However, there is a cancellation among the diagrams for gluino exchange that strongly suppresses their contribution. This was originally shown using a Fierz identity among four-fermion operators [39], but the same result can be obtained with no equations at all. To conserve color, the coefficients of the operators $(QQQL)_{\mathcal{F}}$ and $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ must be totally antisymmetric in the colors of the three quark or antiquark superfields, and since these superfields are bosonic, they must then also be antisymmetric in their flavors as well. Gluino interactions are flavor-independent, so if we can neglect the flavor dependence of the squark masses then the coefficients of the four-fermion $d = 6$ operators will also be totally antisymmetric in flavor as well as color. Fermi statistics then requires the coefficients of these operators to be also totally antisymmetric in the spin indices of the quark or antiquark fields. But the three quark or antiquark fields in the $d = 6$ operators derived by gaugino exchange from $(QQQL)_{\mathcal{F}}$ or $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ are all left-handed and therefore have only two independent spin indices, so no coefficient can be antisymmetric in all three spins. The contribution of gluino exchange to the $d = 6$ operators would therefore vanish if the squark masses were all equal and so this contribution is suppressed by the fractional differences among the masses of the different squarks. According to Eq. (28.6.4), in theories of gauge-mediated supersymmetry breaking the fractional differences between the $\bar{U}$ and $\bar{D}$ squark masses are of order $g'^4/g_s^4$, so gluino exchange between the antisquarks in the $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ operator generates a dimension six four-fermion operator with a coefficient of the same order as that generated by bino exchange. However, since gluinos conserve flavor this operator like the $(\bar{U}\bar{U}\bar{D}\bar{E})_{\mathcal{F}}$ operator must be totally antisymmetric in antiquark flavors, so that it must involve $c$ or $t$ quarks and therefore cannot contribute directly to proton or bound neutron decay. On the other hand, Eq. (28.6.4) indicates that the fractional mass differences among the $Q$ squarks of different flavors is much less than of order $g^4/g_s^4$, so that gluino exchange between the squarks in the $(QQQL)_{\mathcal{F}}$ operator makes a contribution to $g_5$ that is much smaller than wino or bino exchange. We conclude that at least in the theories of gauge-mediated supersymmetry breaking, gluino exchange makes a smaller contribution to proton or bound neutron decay than wino or bino exchange. In other models gluino exchange may make a contribution comparable to that of other processes [40].

For $g \approx g'$ and $m_{\text{wino}} \approx m_{\text{bino}}$, the contribution of wino or bino exchange to the dimension six operators is of order

$$g_6 \approx \frac{g^2 g_5 m_{\text{wino}}}{8\pi^2 m_{\text{squark}}^2}\,, \tag{28.7.4}$$

where $g_5$ is a typical value of the couplings of the effective $d = 5$ interactions (28.7.3). If the wino and squark masses have the same ratio $(g^2/g_s^2)$ as in theories of gauge-mediated supersymmetry breaking, then this gives

$$g_6 \approx \frac{g^4 g_5}{8\pi^2 g_s^2 m_{\text{squark}}}\,. \tag{28.7.5}$$

The four-fermion $qq\ell$-terms of dimensionality six in the effective Lagrangian are the same as those that had been supposed to generate processes like proton decay in non-supersymmetric theories [36]. They produce proton and bound neutron decay at a rate which on dimensional grounds must be of the form

$$\Gamma_N = c_N m_N^5 g_6^2\,, \tag{28.7.6}$$

where $c_N$ is a pure number that must be calculated by non-perturbative calculations in quantum chromodynamics. Much work has gone into these calculations, with results [41] generally in the range $c_N \approx 3 \times 10^{-3\pm 0.7}$.

To estimate $g_5$, we note that it is not possible to produce $\mathcal{F}$-terms like (28.7.3) that involve only left-chiral superfields by the tree-approximation exchange of gauge supermultiplets, which always interact with both left-chiral superfields and their right-chiral complex conjugates. Thus the interactions (28.7.3) arise in the tree approximation only from the exchange of particles of chiral superfields, and therefore $g_5$ is of the order of $g_T^2/M_T$, where $g_T$ is a typical baryon and lepton non-conserving coupling of some superheavy left-chiral superfield of mass $M_T$ to the quark and lepton superfields. To produce the interactions (28.7.3), these superheavy particles must be color triplets or antitriplets, and $SU(2)$ triplets or singlets. Whatever gauge group unifies the strong and electroweak interactions presumably dictates some relation between the interactions of the superheavy color triplet $T$ and the familiar color singlet $H_1$ and $H_2$. Then $g_T$ will be of the same order as the Yukawa coupling constants in the interactions (28.1.2) and (28.1.3) that give mass to the known quarks and leptons, and which are equal to quark or lepton masses divided by the vacuum expectation values of order $G_F^{-1/2} \simeq 300$ GeV of $\mathcal{H}_1^0$ or $\mathcal{H}_2^0$. We therefore take

$$g_5 \approx \frac{G_F m_f^2}{M_T}\,, \tag{28.7.7}$$

where $m_f$ is some typical quark or lepton mass. We have seen that the dimension five operators are antisymmetric in quark flavors, so as a compromise between the masses of the $s$ quark and the $u$ or $d$ quarks, we will take $m_f = 30$ MeV. Combining Eqs. (28.7.5)–(28.7.7) and taking $M_T = 2 \times 10^{16}$ GeV (as suggested by the results of Section 28.2), $c_N = 0.003$, $g_s^2/4\pi = 0.118$, $g^2/4\pi = 1/(0.23 \times 137)$, and $m_{\text{squark}} = 1$ TeV, we find a proton (or bound neutron) lifetime $\Gamma_N^{-1}$ of about $2 \times 10^{31}$ years [42]. This is not very different from experimental lower bounds on the partial lifetimes of what are expected to be the leading modes of proton decay, variously quoted as ranging from $10^{31}$ to $5 \times 10^{32}$ years. At the time of writing the most stringent bounds are set by the non-observation of proton decay in the large Super Kamiokande neutrino detector in Japan [42a]: the partial lifetimes for the decays $p \longrightarrow e^+\pi^0$ and $p \longrightarrow \bar{\nu}K^+$ are greater than $2.1 \times 10^{33}$ years and $5.5 \times 10^{32}$ years, respectively. There is an uncertainty of a factor of at least 100 in the above estimate of the theoretical lifetime from the uncertainty in the squark mass alone, so it is too soon to say that there is any discrepancy between experiment and theoretical expectations. On the other hand, supersymmetry raises the possibility that baryon non-conservation may be discovered soon.

We can also say a little on general grounds about the expected branching ratios for various proton and bound neutron decay modes. As we have mentioned, the dimension five operators (28.7.3) must be totally antisymmetric in the flavor of the quark superfields, so the only operators that concern us here are of the forms $(U_i D_j D_k N_\ell)_\mathcal{F}$, $(D_i U_j U_k E_\ell)_\mathcal{F}$, and $(\bar{D}_i \bar{U}_j \bar{U}_k \bar{E}_\ell)_\mathcal{F}$, where $i, j, k, \ell$ are generation indices, and in each case $j \neq k$. The exchange of neutral winos or binos then produces $d = 6$ four-fermion operators of the forms $u_i d_j d_k \nu_\ell$, $d_i u_j u_k e_\ell$, and $\bar{d}_i \bar{u}_j \bar{u}_k \bar{e}_\ell$, with $j \neq k$ and $i$ and $\ell$ arbitrary, while charged wino exchange produces the same four-fermion operators with $i \neq j$ and $k$ and $\ell$ arbitrary. The only quarks light enough to be involved in proton decay are $u$, $s$, and $d$; ignoring all the others and the small mixing angles in the third generation, we have

$$u_1 = u\,, \qquad d_1 = d\cos\theta_c + s\sin\theta_c\,, \qquad d_2 = -d\sin\theta_c + s\cos\theta_c\,,$$

where $\theta_c$ is the Cabibbo angle, while $u_2$, $u_3$, and $d_3$ can be ignored. The four-fermion operators that can be produced by wino or bino exchange and can contribute to proton or bound neutron decay thus are $uds\nu_\ell\cos(2\theta_c)$, $udd\nu_\ell\sin(2\theta_c)$, $usue_\ell\cos\theta_c$, and $uude_\ell\sin\theta_c$, plus others with quarks and leptons replaced with antiquarks and antileptons. All other things being equal, the dominant decay modes are therefore $p \longrightarrow K^+\bar\nu$, $n \longrightarrow K^0\bar\nu$, $p \longrightarrow K^0e^+$, and $p \longrightarrow K^0\mu^+$, while the rates for the decay modes $p \longrightarrow \pi^+\bar\nu$, $n \longrightarrow \pi^0\bar\nu$, $p \longrightarrow \pi^0e^+$, $p \longrightarrow \pi^0\mu^+$, and $n \longrightarrow \pi^-e^+$ are suppressed by factors $\sin^2\theta_c = 0.05$, though also enhanced somewhat by the greater phase space available.

These considerations do not lead to definite predictions of branching ratios, because in addition to all the factors mentioned above, the coefficients generically called $g_5$ of the operators (28.7.3) may have a strong dependence on the superfield flavors appearing in these operators. To go further, one needs a specific theory for the generation of the dimension five operators. Most of the authors of Reference 42 concluded on the basis of a supersymmetric version of $SU(5)$ theories that proton and bound neutron decay would be dominated by the processes $p \longrightarrow K^+\bar\nu$ and $n \longrightarrow K^0\bar\nu$, but for a model based on $SO(10)$ charged lepton modes can become prominent [43]. Also, in some models higgsino exchange can compete with wino and bino exchange [44], increasing the rate of $p \longrightarrow K^+\bar\nu$. It seems a good idea in searches for baryon non-conservation to keep an open mind as to the decay modes to be expected in proton or bound neutron decay.

Of course, it is possible that all of these baryon non-conserving processes are prohibited by some sort of conservation law. As mentioned in Section 28.1, string theory argues against baryon conservation being a fundamental global continuous symmetry, but the baryon non-conserving operators (28.7.3) may be forbidden by a $\mathbb{Z}_3$ multiplicative symmetry known as baryon parity [45], under which the $Q$ superfield is neutral; the $H_2$ and $\bar D$ superfields are multiplied by the phase $\exp(i\pi/3)$, and the $L$, $H_1$, $\bar U$, and $\bar E$ superfields are multiplied by the opposite phase $\exp(-i\pi/3)$. This symmetry is designed to allow the fundamental Yukawa couplings (28.1.2) and (28.1.3) as well as the $\mu$-term (28.5.7) and the lepton non-conserving terms (28.1.4) and (28.7.1), but it rules out the dimension four baryon non-conserving term (28.1.5) and the dimension five baryon non-conserving terms (28.7.3). This symmetry is spontaneously broken by the appearance of vacuum expectation values of $\mathcal{H}_1^0$ and $\mathcal{H}_2^0$ (and perhaps the sneutrino fields $\mathcal{N}$), and with no conservation law for $R$ parity, there would be nothing to keep the lightest supersymmetric particle stable.